

## Experiment 9

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
df = sns.load_dataset('titanic')
print(df.head())

print(df.info())
print(df.isnull().sum())

sns.boxplot( y=df['age'] )
plt.title("Boxplot of Age")
plt.show()

sns.boxplot( y=df['fare'] )
plt.title("Boxplot of Fare")
plt.show()

print(df['survived'].value_counts())
sns.countplot( x='survived', data=df )
plt.title("Survival Analysis")
plt.show()

plt.hist(df['age'].dropna())
plt.xlabel("Age")
plt.ylabel("Count")
plt.title("Age Distribution")
plt.show()

Q1 = df['fare'].quantile(0.25)
Q3 = df['fare'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
outliers = df[ (df['fare'] < lower) | (df['fare'] > upper) ]
print(outliers)
```